

Review

X-Ray Characterization of Semiconductor Materials and Advanced Packaging: A Perspective on Multidimensional Structural Analysis

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Abstract

X-ray techniques provide powerful, non-destructive tools for structural characterization in semiconductor manufacturing and advanced packaging. Their strong penetration capability and sensitivity to multiple contrast mechanisms enable the investigation of lattice structure, strain, defects, interfaces, and elemental distribution across a wide range of length scales. As semiconductor devices evolve toward three-dimensional architectures and heterogeneous integration, there is an increasing demand for characterization approaches capable of probing complex, buried, and multi-scale structures in a consistent manner. In this review, we present a systematic overview of X-ray characterization techniques for advanced semiconductor systems, including diffraction-based methods, small-angle scattering, computed tomography, X-ray fluorescence, and spectroscopic approaches. These techniques are discussed in terms of the type of structural, morphological, and compositional information they provide, their applicable length scales, and their strengths and limitations in addressing key challenges such as thin films, high-aspect-ratio structures, buried interfaces, and full wafers. Particular attention is given to the complementary nature of different X-ray modalities and their roles in addressing practical metrology problems. The limitations associated with resolution, model dependence, and data interpretation are also outlined. Finally, emerging opportunities in laboratory X-ray sources, synchrotron-based methods, and integrated characterization strategies are briefly discussed. This review aims to provide a unified perspective for understanding and integrating X-ray techniques, offering insights into their roles in addressing the growing complexity of next-generation semiconductor devices.

Keywords: X-ray diffraction and scattering; X-ray spectroscopy; three-dimensional X-ray imaging; semiconductor characterization; advanced packaging



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1. Introduction

Structural characterization plays a central role in understanding and engineering modern material and device systems. Among the available approaches, X-ray-based techniques provide a uniquely powerful platform due to their strong penetration capability and sensitivity to multiple contrast mechanisms within a unified physical framework. These advantages are becoming increasingly important as semiconductor technologies undergo rapid structural evolution driven by both “More Moore” and “More-than-Moore” paradigms. On the one hand, as device dimensions approach the atomic scale [1], high-aspect-ratio (HAR) structures such as three-dimensional FinFETs, nanosheets, and Gate-all-around (GAA) architectures [2,3], have increased substantially. At the same time, advanced packaging and heterogeneous integration are introducing see-through architectures that integrate multiple chips and materials within three-dimensional configurations. This has led to emerging interconnect structures like through-silicon vias (TSVs), micro-bumps, and hybrid bonding, enabling ultra-short-distance, ultra-high-bandwidth connections between chips [4,5].

As a result, key structural challenges in modern semiconductor systems can be broadly categorized into several recurring types, including thin films and multilayers, high-aspect-ratio nanostructures, buried or see-through architectures, and wafer-scale variations. These challenges span multiple length scales, information dimensions, and collectively determine device performance and reliability. For example, thin-film properties—such as thickness, density, and interface roughness—govern dielectric behavior; geometrical parameters in high-aspect-ratio devices influence electrostatic control and variability; buried interface quality affects interconnect reliability in three-dimensional integration; and wafer-scale non-uniformities impact yield and large-scale device performance.

In response to these challenges, semiconductor metrology is undergoing a paradigm shift: transitioning from two-dimensional geometric measurements to three-dimensional and multi-physics coupled analysis. At the same time, rapid advances in high-brightness synchrotron radiation sources [6], laboratory microfocus X-ray sources, and high-performance detectors have significantly expanded the capabilities and accessibility of X-ray metrology. From this perspective, modern X-ray methods are evolving from isolated characterization tools into a multidimensional crystallographic framework.

Despite the rapid development of these methods, existing reviews are often organized around individual techniques, making it difficult to obtain a unified understanding of how different X-ray approaches relate to specific semiconductor characterization challenges. In practical scenarios, multiple techniques are often required to fully resolve complex structures, particularly in systems involving multi-scale features and heterogeneous integration. In particular, as shown in Table 1, the same practical challenge often requires multiple information dimensions. Accordingly, this review aims to provide a structured perspective on X-ray characterization techniques by organizing them according to the types of information they provide across different dimensions. The relationships and potential complementarity between these techniques are discussed where appropriate.

Table 1. Representative X-ray techniques for key challenges in semiconductor manufacturing and advanced packaging.

Information	Challenges	Techniques	Film	High-Aspect-Ratio	See-Through/Backside-Relevant Inspection	Wafer/Die
	Structural Characterization	XRD-based, SAXS-based, XRR	XRD for multilayer strain and crystalline quality; GISAXS for periodic morphology of nanostructured films; XRR for thickness/roughness	SAXS for profile, tilt, sidewall reconstruction, and LWR/LER metrology;	μ-XRD for buried strain fields and buried crystalline distortion	μ-XRD and XRDI for strain, warpage-related stress, and die-level crystallographic non-uniformity
	Morphology Characterization	CT/CL, TXM, PXCT	Limited role	CT/CL trench, hole, edge, and via geometry; PXCT for nanoscale 3D reconstruction	CT/TXM/PXCT for buried cavities, residues, voids, TSVs, hybrid-bond defects, post-thinning defects, and backside metal inspection	CT/CL mainly for post-bonding internal defects rather than primary geometry metrology
	Compositional Characterization	XRF-based, XPS-based	GIXRF for elemental depth profiles; HAXPES/SW-XPS for interface chemistry and band alignment	GIXRF or TXRF for buried coatings or composition-sensitive cases	HAXPES for buried interfaces and residues; XRF-based approaches for buried elemental heterogeneity	TXRF for wafer contamination; full-field XRF for large-area elemental non-uniformity

2. X-Ray Scattering and Diffraction Techniques for Microstructural Characterization

In modern semiconductor devices, many of the most important structural variables are not directly visible as geometric objects, but manifest as changes in lattice spacing, crystal tilt, strain partitioning, and periodic electron-density modulation. These quantities govern channel mobility, epitaxial stability, stress engineering, and process-induced relaxation, yet they are often buried inside multilayer stacks and cannot be quantitatively obtained by surface-sensitive inspection or direct cross-sectional imaging alone. Physically, because these problems originate from periodic atomic order, they are most naturally encoded in reciprocal space through the positions, widths, asymmetries, and intensity distributions of diffraction or scattering signals. For this reason, X-ray diffraction, scattering, and reflectivity are indispensable when the scientific question is how the lattice is distorted, how periodic nanostructures are formed, or how buried crystalline order evolves in semiconductor structures.

2.1. X-Ray Diffraction (XRD) for Crystal Orientation and Strain Metrology

Precise control of crystal orientation and lattice strain is fundamental to semiconductor manufacturing, as these parameters govern epitaxial growth behavior, strain engineering, and device performance. X-ray diffraction (XRD) provides a direct and quantitative approach for measuring lattice parameters, crystal orientation, and strain states in semiconductor structures. From an information perspective, XRD provides both statistically averaged structural data and spatially resolved insights. As device architecture evolves toward three-dimensional configurations and localized strain engineering, XRD has progressively expanded from conventional two-dimensional statistical characterization to quasi-three-dimensional approaches with limited depth sensitivity, and, more recently, to truly three-dimensional diffraction-based structural characterization.

Traditional XRD employs a spot size ranging from millimeters to hundreds of micrometers to illuminate the sample. The detector scans through angular space to acquire diffraction signals, with the resulting measurements essentially representing the statistical average response of the crystal structure within the spot's coverage area. Consequently, statistical XRD does not provide spatial distribution information but instead yields macroscopic quantities such as average lattice parameters, orientation dispersion, and defect fluctuations. Two core types of information can be extracted from these diffraction peaks. First, the position of each peak directly corresponds to the interplanar spacing of a specific crystal plane. Precise measurement of minute shifts in peak positions enables calculation of elastic strain in the lattice under applied force. For example, in SiGe strain channels, epitaxial layer-matching structures, and metal interconnect layers, measurements of symmetric and asymmetric reflections can, respectively, extract the in-plane and out-of-plane strain components, enabling evaluation of epitaxial quality and residual stress states [7–10]. For example, the systematic shift of the SiGe diffraction peak with decreasing fin width provides direct evidence of strain relaxation at the nanoscale, demonstrating the sensitivity of XRD to process-induced dimensional effects. Second, when grain size decreases, or micro-strain arises from dislocations in the lattice, the width of diffraction peaks increases accordingly. The half-width of the rocking curve, obtained by slowly oscillating the sample at a fixed diffraction angle, directly quantifies the degree of deviation from the ideal texture: the more dispersed the orientation, the broader the curve. In metal interconnect layers, grain orientation directly impacts electromigration lifetime. For instance, in Al interconnect films, the (111) preferred orientation reduces the number of grain-boundary diffusion pathways, thereby enhancing reliability. Precise measurement of swing curve width enables quantitative comparison of how different underlayers influence texture, significantly

delivering an order-of-magnitude improvement in electromigration lifetime [11]. In copper interconnects, XRD polar pattern analysis reveals correlations between fiber texture and plating additives/annealing processes, providing a basis for optimizing fill quality and suppressing void formation [12–14]. This demonstrates that statistically oriented information measured by XRD can be directly converted into engineering reliability metrics.

In integrated circuits manufacturing, XRD has primarily evolved along three experimental systems: monochromatic microbeam diffraction, white-light Laue diffraction, and X-ray diffraction imaging (XRFI). Despite differing experimental geometries and data formats, these techniques have evolved methodologically from two-dimensional localized scanning to quasi-three-dimensional depth analysis, providing critical support for semiconductor process optimization and reliability assessment.

Monochromatic scanning diffraction employs a focused monochromatic X-ray beam to scan the sample point by point, achieving high spatial resolution and extensive detection coverage across different scales, such as structural imaging of wafers [15], warpage analysis of packaged chips [16], and characterization of localized strain in advanced node devices such as FinFETs [17]. Building upon this, Shida et al. developed a quasi-3D tomographic mapping method by exploiting the continuous depth variation of Ge content in SiGe buffer layers and its correlation with the Bragg angle (2θ). Because the Ge composition varies monotonically with depth, the corresponding change in lattice spacing leads to a depth-dependent shift in 2θ , allowing the 2θ dimension in the 3D reciprocal space to be used as a proxy for depth. This enables quasi-3D reconstruction of the distribution of mismatch dislocations in a non-destructive manner [18]. When the focus shifts to polycrystalline metal interconnects, white-light Laue diffraction is typically employed, enabling each grain to generate multiple sets of diffraction spots under fixed-geometry conditions. It is currently widely used to address stress concentration issues caused by thermal expansion mismatch in TSV copper fillings within 3D packaging, as well as to identify regions of localized elastic strain and plastic deformation [19–21]. Similarly, combined with differential probe technology, it achieves the first non-destructive quantitative measurement of the full stress tensor at varying depths within TSVs [19]. X-ray diffraction imaging techniques capture two-dimensional projections of lattice distortions using wide beams, followed by B-spline surface fitting [22,23] and energy-resolved slicing [24] to yield three-dimensional distributions of silicon lattice bending surfaces in packaged chips and minute lattice tilts ($\pm 0.03^\circ$).

Building upon the aforementioned quasi-3D techniques, a new generation of methods capable of directly acquiring three-dimensional volumetric information has emerged in recent years. These technologies no longer rely solely on integration or tomographic mapping in the depth direction. Instead, they achieve three-dimensional reconstruction of strain fields within individual grains or entire polycrystalline networks through coherent diffraction or tomographic geometries. Bragg coherent diffraction imaging (BCDI) directly reconstructs the three-dimensional strain and displacement fields within individual grains or nanostructures by inverting three-dimensional coherent diffraction patterns collected around one or more Bragg diffraction peaks. This technique has been successfully applied to map nanoscale strain distributions in silicon-on-insulator (SOI) nanostructures [25]. The introduction of scanning redundancy concepts has led to the development of Bragg ptychography, which significantly enhances the resolution and scanning range of BCDI technology. This advancement enables the detection of nanoscale strain and defect distributions in electronic devices and chips within wafer-level packaging [26–28]. Besides the above technologies, diffraction contrast tomography (DCT) [29] and three-dimensional X-ray diffraction (3DXRD) [30,31] aim to resolve the three-dimensional orientation and strain networks of hundreds to thousands of grains in polycrystalline materials. With

breakthroughs in laboratory light-source technology [32], these techniques will gradually demonstrate significant engineering potential for reliability studies of semiconductor interconnect metals and power-device metallization layers.

Overall, diffraction-based X-ray methods reconstruct the various crystallographic quantities. Table 2 compares several representative approaches in terms of their most relevant semiconductor applications, strengths, and limitations.

Table 2. Comparison of X-ray diffraction techniques for strain and defect characterization.

Techniques	Main Advantages	Main Limitations	Typical Applications
Statistics XRD	High accuracy, reliable quantification	No spatial distribution information	Mainly Film
Monochromatic Microbeam Scanning XRD	High spatial resolution, capable of mapping	Time-consuming scanning, limited depth information	Film, wafers, and see-through structures
White Beam Laue Diffraction	Multiple diffraction spots in single exposure, suitable for polycrystals	Complex data analysis	HAR and see-through structures
XRDI	Large area, fast imaging	Limited quantification capability	Wafers and device-scale
BCDI and Bragg Ptychography	Nanoscale resolution, direct reconstruction of displacement fields	Requires coherent source, demanding sample requirements	Buried nanostructures
3DXRD/DCT	Strong statistical capability for polycrystals	Lower resolution compared to coherent methods	Polycrystalline metal layers

2.2. Small-Angle X-Ray Scattering (SAXS) for Critical Dimension Metrology

The transition toward increasingly complex three-dimensional device architectures has made accurate characterization of nanoscale feature sizes and periodic structures more challenging. Critical dimension metrology must now capture structural information from densely packed features with high statistical reliability. Small-angle X-ray scattering (SAXS) probes the spatial correlations of nanoscale structures [32] by analyzing the angular distribution of X-rays scattered at small angles. Because geometric parameters such as feature size, pitch, and edge roughness directly modulate the scattering intensity and reciprocal-space distribution, SAXS enables the quantitative extraction of critical dimensions (CDs), line-edge roughness, and average cross-sectional profiles in periodic semiconductor nanostructures [33–35]. The fundamental principle involves irradiating the sample with a collimated, monochromatic X-ray beam and recording coherent scattering signals at small scattering angles, which originate from nanoscale inhomogeneities in electron density.

The quantitative relationship between intensity $I(q)$ and scattering vector q , derived from scattering theory, form the core of SAXS analysis:

$$I(q) \propto |F(q)|^2 = |\Delta\rho|^2 V^2 P(q) S(q) \quad (1)$$

where $\Delta\rho$ is the difference in electron density between the scatterer and the matrix, V is the scatterer volume, $P(q)$ is the shape factor reflecting the geometric characteristics of a single particle, and $S(q)$ is the structure factor describing the spatial arrangement and interactions between particles.

In advanced semiconductor devices, photolithographic patterns such as line grids and through-hole arrays are considered highly periodic grating structures. CD-SAXS essentially

measures the Fourier distribution $P(q)$ of the structure's electron density. The positions of diffraction peaks determine the structural periodicity, while the intensity distribution across different orders encodes information about the cross-sectional morphology within a single period. For high-aspect-ratio structures such as 3D NAND channel holes, the scattering amplitude can be interpreted as the coherent superposition of Fourier components from different depth directions. Structural variations along the sidewalls, therefore, modulate the intensity distribution of higher diffraction orders. For example, sidewall curvature alters the intensity envelope of diffraction peaks, while bottom tilts introduce peak asymmetry and broadening. In high-aspect-ratio structures, the scattering amplitude approximately scales with the structure height, while the scattered intensity scales with the square of the height, providing a significant signal advantage for CD-SAXS in high-aspect-ratio semiconductor architectures.

Based on this scattering framework, structural information is distributed across different regions of reciprocal space. In practical CD-SAXS analysis, different q ranges encode structural features at distinct length scales, enabling hierarchical reconstruction of semiconductor nanostructures: the low- q region primarily encodes overall periodicity and macroscopic geometric perturbations, making it particularly critical for inline CD control during back-end-of-line (BEOL) processes [36,37]. The middle- q region contains detailed information on sidewall curvature and profile evolution and plays a central role in enabling model-based reconstruction of three-dimensional structural profiles, such as “two-bow” or multi-curvature structures. By jointly fitting multi-order scattering intensity distributions, stable reconstruction of sidewall profiles along the depth direction is achieved. CD-SAXS measurements performed on semiconductor wafers show excellent agreement with cross-sectional TEM observations, confirming that the mid- q region of the scattering spectrum contains sufficient information to constrain three-dimensional profile reconstruction [38]. The high q region is sensitive to nanoscale interfacial structures and thin layers. As demonstrated by Rigaku team, a ~2 nm thick sidewall metal coating produces detectable scattering contributions due to its strong electron density gradient, enabling quantitative analysis of deeply buried thin films. These capabilities demonstrate that CD-SAXS provides a unified metrology framework capable of simultaneously resolving macroscopic structural geometry and nanoscale interfacial features in advanced semiconductor architectures [39].

However, this effective utilization of the medium-to-high q region imposes more stringent requirements on experimental conditions. Early high-precision SAXS studies relied almost entirely on synchrotron radiation sources [40]. For transmission CD-SAXS, which requires penetration through 750 μm silicon wafers, X-ray energies exceeding 15 keV are necessary to minimize absorption losses. Over the past two decades, laboratory source technologies such as micro-focused rotating anodes and liquid metal jets have significantly enhanced brightness, enabling high-energy laboratory systems to gradually support CD-SAXS. However, in online metrology scenarios for logic devices, achieving sub-minute measurement throughput with a spot size under 100 μm while maintaining a divergence angle below the diffraction-peak spacing of 1 mrad poses a challenge that requires at least a two-order-of-magnitude increase in source brightness [41]. In contrast, memory devices have become the first commercial field for CD-SAXS online applications due to their strong scattering signals from high-aspect-ratio structures. Multiple equipment suppliers have launched compact transmission SAXS systems suitable for cleanrooms, such as the Bruker Sirius-XCD, which can complete single-point measurements within minutes [42]. Moreover, the true expansion of CD-SAXS capabilities stems not only from improvements in light-source and detector performance but also from continuous optimization of the inverse-problem-solving framework [35,43]. Traditional iterative approximation methods struggle to meet the high-throughput demands of production lines, but the application

of deep learning is gradually changing this landscape. Through convolutional neural networks [44], ResNet [45], and other architectures, analysis time can be compressed from hours to seconds.

Although transmission CD-SAXS is well suited for deep structures, its strong penetration limits sensitivity to surface features such as interface roughness, line edge roughness, and shallow buried structures. This limitation arises from the measurement geometry itself, as transmission scattering primarily captures structural information integrated along the beam path. To enhance sensitivity to surface and near-surface morphology, the scattering geometry can instead be configured in grazing-incidence mode (GI-SAXS), where the incident angle approaches the total internal reflection critical angle. Under these conditions, the scattering pattern forms truncation rods along the vertical direction, which encode information about vertical topographical and surface morphology.

For HAR structures, the key metrology question is not simply whether the feature can be imaged, but whether its buried profile can be reconstructed with adequate depth sensitivity and statistical process relevance. Table 3 compares CD-SAXS with several representative alternatives from this perspective.

Table 3. Comparison of methods for HAR profile metrology.

Technique	Advantages	Limitations
CD-SAXS (Transmission)	Highly sensitive to HAR structures	Requires high throughput
GI-SAXS	Sensitive to surface/interface & LER/LWR	Limited depth sensitivity; complex modeling
SEM	Fast top-view CD & edge imaging	Mainly surface sensitivity
TEM (cross-section)	Direct real-space structure	Destructive, limited statistics
AFM/3D-AFM	High-resolution surface topography & roughness	Limited depth sensitivity; slow
X-ray CT	3D density distribution	Resolution insufficient for nanoscale CD

Compared with CD-SAXS, GI-SAXS does not require X-ray penetration through thick substrates, enabling high-precision measurements under lower-energy laboratory light sources. However, this geometry also introduces additional modeling complexity, particularly when multiple scattering and significant interface roughness are present. Despite these challenges, GI-SAXS has become an effective tool for quantitative characterization of nanoscale surface structures in advanced semiconductor processes [46–48]. For example, Freychet et al. employed a conventional Cu-K α laboratory source to measure etched silicon line gratings under grazing-incidence geometry. By rotating the sample to acquire angle-resolved datasets and applying inverse reconstruction using the CMA-ES algorithm, they obtained angle-resolved datasets and successfully reconstructed the depth profile of the line structures. Results demonstrated high agreement between the GI-SAXS-extracted profile and 3D-AFM measurements. The reconstructed profiles showed strong agreement with independent measurements obtained from 3D-AFM [48]. In addition, the development of high-brightness liquid-metal jet X-ray sources has further facilitated the implementation of laboratory-scale GI-SAXS measurements, enabling broader adoption of this technique for semiconductor nanostructure characterizations [49].

Different metrology techniques address distinct aspects of HAR structures, including deterministic geometrical parameters (e.g., CD and profile) and stochastic features (e.g., LER/LWR). These measurement tasks are implicitly reflected in the strengths and limitations of each technique. Table 3 compares SAXS-based techniques with several representative alternatives from this perspective.

The unique value of SAXS, therefore, lies not only in its non-destructive character, but in its sensitivity to the full buried profile of periodic HAR structures, where both direct cross-sectional methods face important trade-offs.

2.3. X-Ray Reflectivity (XRR) for Thin-Film Thickness, Density, and Interface Characterization

Thin films form the fundamental building blocks of modern semiconductor devices, including gate stacks, dielectric layers, barrier layers, and multilayer interconnect structures. Precise characterization of thin-film thickness, density, and interface quality is therefore essential for ensuring device performance and process reliability, making X-ray reflectivity (XRR) an important metrology technique for thin-film analysis. XRR is a non-destructive characterization technique based on total internal reflection and multi-layer interference effects of X-rays at smooth interfaces, widely used for precise analysis of thin films and multilayer structures [50–52]. From a scattering theory perspective, XRR represents a special case where the parallel component of the scattering vector is zero. Under these conditions, scattering is determined solely by variations in electron density along the surface normal direction. Although both XRR and GI-SAXS employ similar low-incidence-angle geometries, they differ in the dimensionality of the information they capture. XRR records only specular reflection signals and exhibits high sensitivity to one-dimensional layered structures perpendicular to the surface. It is primarily used for precise measurements of average film thickness, electron density, and interfacial roughness. In contrast, GI-SAXS records two-dimensional scattering distributions that simultaneously incorporate in-plane and normal-related information. It is more suitable for analyzing lateral correlation lengths, periodic structures, and statistical roughness topography. Thus, the two techniques form a complementary relationship in thin-film structure characterization.

The physical basis of XRR can be equivalently established on the complex refractive index of materials to X-rays:

$$n = 1 - \delta - i\beta \quad (2)$$

where δ and β are material-dependent parameters related to the electron density and absorption coefficient, respectively. The reflectivity profile—measured as a function of the normal momentum transfer q_z —contains rich structural information about layered thin films. By analyzing the variation in reflection intensity with q_z , several key structural parameters of multilayer thin films can be extracted by fitting the measured reflectivity curve, most notably the film density, thickness, and interface roughness. Film density derived from XRR is frequently used to evaluate film compactness, porosity, and the impact of deposition processes on material quality for both low- k dielectric and high- k gate dielectric films [50,53–55]. In a multilayer thin-film structure, interference between reflected waves at adjacent interfaces in multilayer structures generates Kiessig fringes on the reflectance curve. Their period Δq_z approximates the film thickness as follows:

$$d \approx \frac{2\pi}{\Delta q_z} \quad (3)$$

This relationship enables XRR to be widely used in atomic layer deposition (ALD) processes to precisely determine the growth per cycle and verify process linearity and repeatability [56,57]. In addition, interface or surface roughness induces overall attenuation of reflected signals in high q_z regions, an effect commonly described using Debye–Waller-like factors. Such roughness characterization is particularly important for advanced semiconductor heterostructures. For instance, in advanced logic devices, XRR provides critical insights into the interface roughness of SiGe/Si superlattices within complementary field-effect transistors (CFETs) [54].

XRR plays an important role in evaluating the quality of gate dielectrics in advanced semiconductor devices. For example, in the structural characterization of PEALD HfO₂ gate dielectrics for GaN-based power devices [58], fitting results (see Figure 1) revealed clear Kiessig oscillations and quantified an electron density of

9.1 g/cm³, approaching the bulk limit. At the same time, the HfO₂/GaN interface roughness was determined to be approximately 0.5 nm. These structural parameters provide direct physical evidence explaining the high dielectric constant and high breakdown field strength of the device.

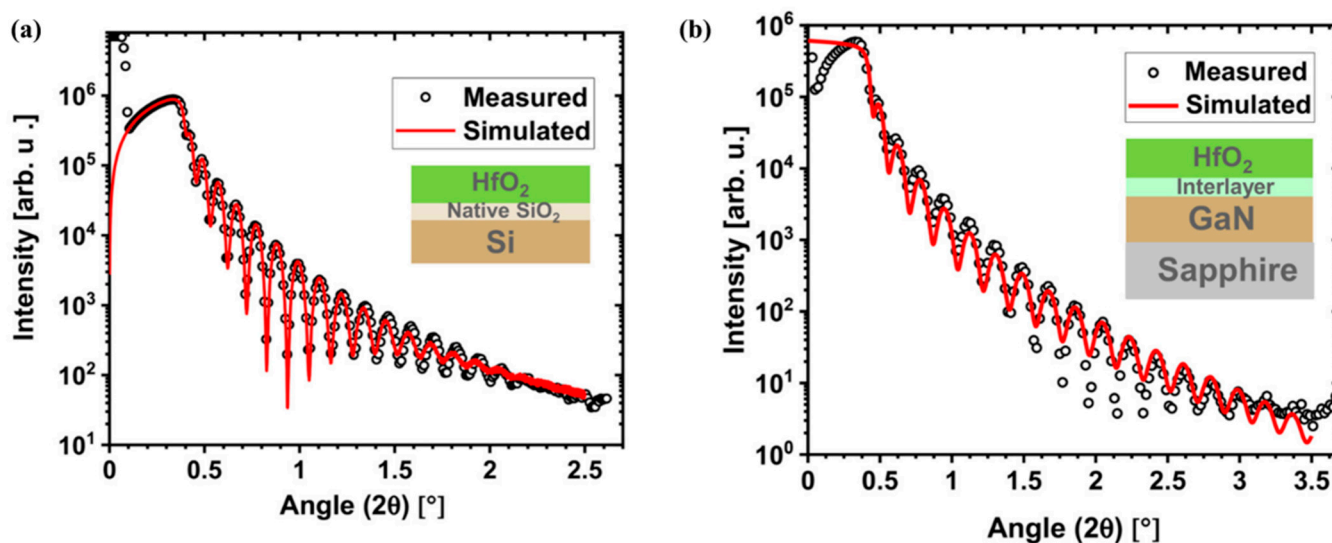


Figure 1. XRR analysis of ~25 nm PEALD HfO₂ films. Measured and simulated XRR angular scans on (a) Si and (b) GaN [58]. Adapted from Paul, P.; Brusaterra, E., *J. Vac. Sci. Technol. A* (2025), under the Creative Commons Attribution (CC BY) license.

XRR has also been applied to investigate structure–property correlations in resistive random-access memory devices [59]. As shown in Figure 2, XRR curves of TiO_{2–x}/Pt heterostructures exhibit systematic shifts in the critical angle with increasing irradiation dose, indicating variations in the average electron density of the film. By reconstructing the electron density profiles $\rho(z)$, a surface-enriched titanium layer and a subsurface region of reduced electron density were identified. This electron density gradient directly correlates with oxygen ion migration, providing a structural basis for conductive filament formation.

As device structures become increasingly complex, traditional XRR fitting methods based on manual model adjustments have revealed issues of computational time consumption and non-unique solutions. In recent years, the integration of machine learning with physical scattering models has emerged as an effective strategy to overcome these limitations. On the one hand, such approaches can directly map XRR data to electrical parameters, enabling end-to-end device performance prediction and process defect detection [60]. On the other hand, reliable structural parameters can be consistently obtained even under conditions of significantly reduced sampling or radiation dose [61–63]. These developments enable XRR to evolve from an offline analysis technique into an in-line metrology tool capable of real-time process monitoring.

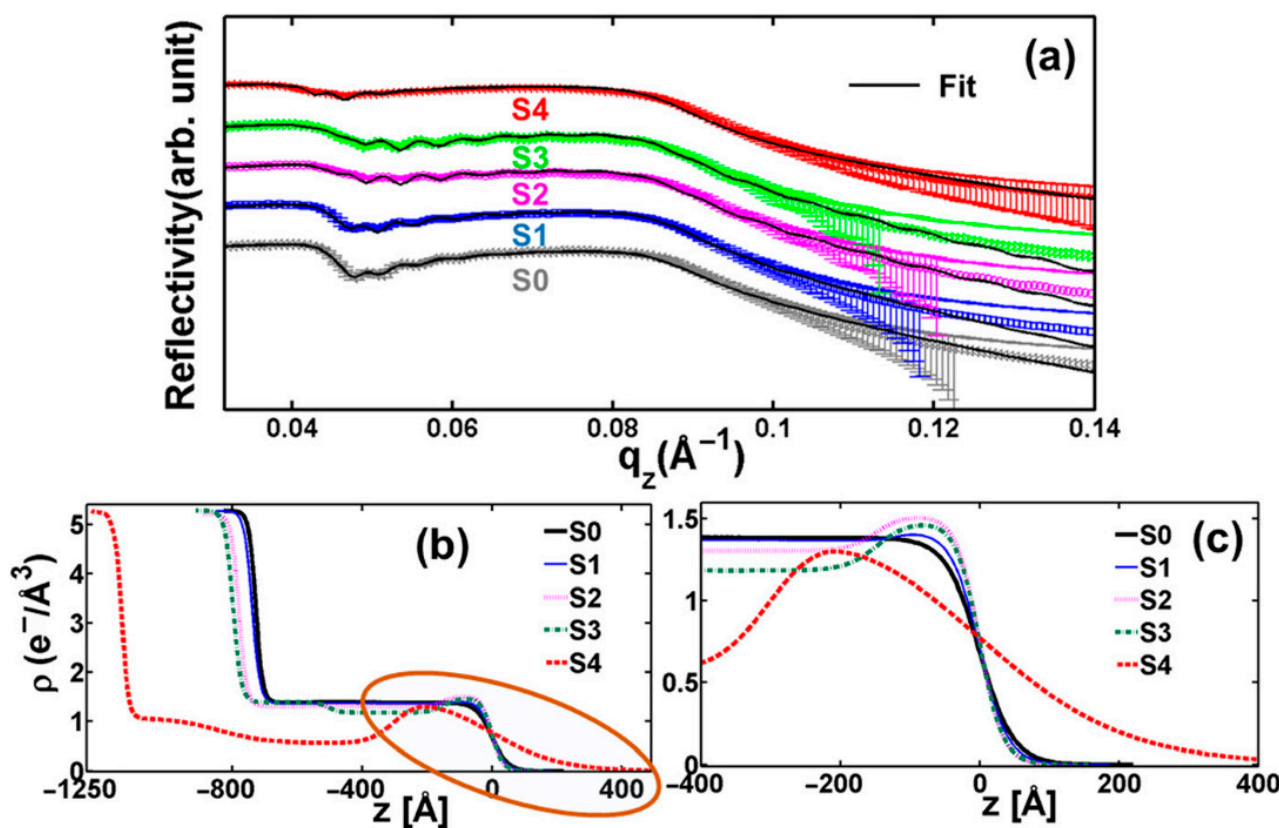


Figure 2. XRR investigation of irradiation-induced structural changes in $\text{TiO}_{2-x}/\text{Pt}$ heterostructures. (a) Reflectivity curves shift systematically with increasing Ar^+ fluence, indicating variations in average electron density. (b) Reconstructed electron density profiles reveal a surface-enriched Ti layer and a subsurface region of reduced density, attributed to oxygen vacancy formation. (c) Magnified view of the near-surface region (circled (saffron colored) area in (b)). The results establish a direct link between oxygen deficiency and conductive filament formation in resistive switching devices [59]. Reprinted with permission from Barman and Saini, Appl. Phys. Lett. 108, 244101 (2016). Copyright 2016 AIP Publishing LLC.

3. X-Ray Tomography for See-Through Structures and Defects

However, semiconductor failure and structural evolution are not determined by lattice metrics alone. In advanced packaging, three-dimensional interconnects, and heterogeneous integration, practical problems often arise from voids, cracks, delamination, sidewall deformation, or local geometric non-uniformity. These defects are fundamentally real-space objects: what matters is their position, connectivity, morphology, and three-dimensional distribution inside the device. Although reciprocal-space methods are highly sensitive to structural perturbation, they often do not uniquely specify the actual geometry of buried defects or internal architectures. This is why real-space X-ray imaging and tomography become necessary: they provide direct non-destructive reconstruction of internal morphology when the scientific question concerns where a defect is, how it propagates, and how three-dimensional device geometry evolves.

3.1. X-Ray Microscopy for 3D Packaging Defect Inspection

High-resolution X-ray microscopy offers unique capabilities for non-destructive characterization of semiconductor devices across a broad range of length scales from micrometers to nanometers [64,65]. Due to its robustness and compatibility with relatively large sample volumes, conventional X-ray computed tomography (XCT) and micro-XCT are widely used for defect detection and failure localization. As shown in Figure 3a,b, in XCT, a parallel

or cone-shaped X-ray beam is transmitted through the sample while it is rotated over a full angular range, and a series of absorption-based projection images are collected. A three-dimensional volume is then reconstructed using back-projection algorithms from these projections. However, their spatial resolution is typically limited to $\sim 0.5\ \mu\text{m}$ when relying solely on geometric magnification.

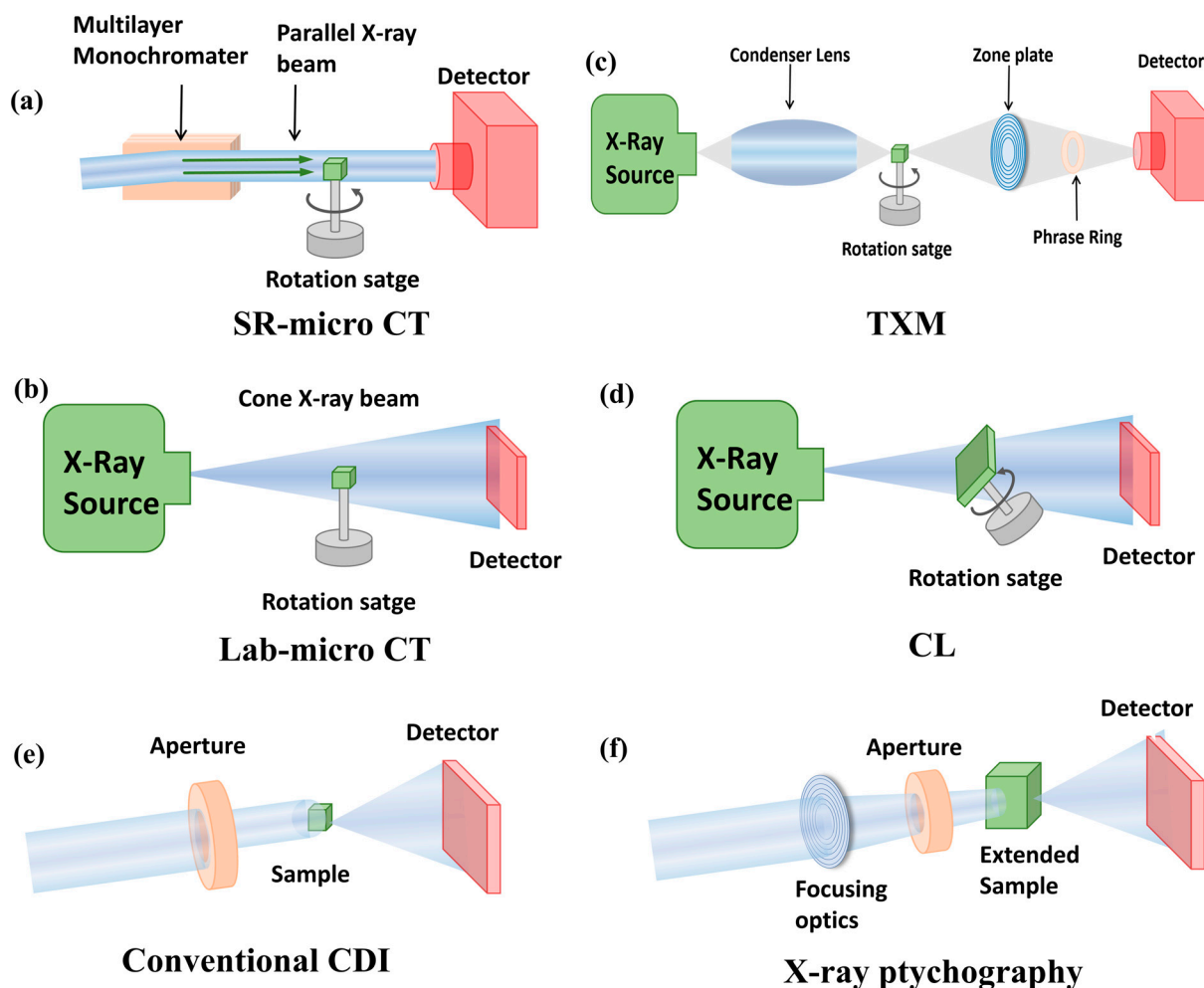


Figure 3. Several typical configurations of X-ray imaging. (a) SR-micro CT. (b) Lab-micro CT. (c) TXM. (d) CL. (e) CDI. (f) ptychography. The green arrow indicates parallel X-ray beam.

To overcome this limitation, full-field transmission X-ray microscopy (TXM) has been developed. In TXM, a focused X-ray beam is used together with X-ray optics, such as zone plates, to achieve higher resolution projection images of the sample. In addition to absorption contrast, phase contrast can be achieved either through propagation-based methods or by introducing a phase ring, enabling enhanced visualization of weakly absorbing structures (see Figure 3c). In synchrotron-based TXM systems, the combination of high-brightness X-rays, high-resolution focusing optics, and efficient CMOS area detectors [66] enables sub-10-nanometer spatial resolution in the soft X-ray region [67,68] and tens-of-nanometer spatial resolution in the hard X-ray region [69,70]. For laboratory instruments, advanced nano-focus X-ray sources currently achieve a spatial resolution of approximately 150 nm [71]. Nevertheless, rotating anode sources remain the dominant laboratory solution because of their stability, reliability, and mature system integration, typically enabling around 50 nm resolution imaging in commercial systems.

Concurrently, liquid metal jet (LMJ) sources [72–74] have been introduced into commercial XRM platforms in recent years, offering significantly higher brightness and broader

spectral flexibility. Although LMJ sources have also been employed to explore laboratory ptychography [75], the spatial resolution currently achieved remains below that of state-of-the-art TXM systems and far from the performance of synchrotron ptychography. Consequently, rotating-anode-based X-ray microscopes still dominate the advanced packaging inspection in laboratory environments. Focusing optics represent another core component of TXM systems. Common optical elements include Kirkpatrick–Baez (KB) mirrors, Fresnel zone plates (FZPs), composite refractive lenses (CRLs), and multilayer film lenses [73], each presenting distinct trade-offs between numerical aperture, efficiency, and robustness. Among these, Fresnel zone plates are most widely adopted in commercial laboratory TXM systems due to their relative simplicity of alignment and operational stability. With continued progress in nanofabrication, zone plates with outermost ring widths below 30 nanometers and high aspect ratios can be fabricated, pushing the spatial resolution of hard X-ray microscopy close to practical limits [76,77].

Traditional XCT primarily relies on absorption contrast. However, at higher photon energies, the attenuation contrast between materials decreases, which significantly reduces image quality. This effect is particularly pronounced in advanced semiconductor packaging, where copper interconnects are embedded within low-dielectric-constant materials, and phase effects often dominate imaging in the hard X-ray regime. Consequently, modern TXM systems predominantly employ phase-contrast imaging modes. Among various phase-contrast methods, Zernike phase-contrast microscopy has become one of the most widely adopted techniques in laboratory TXM. By introducing a phase ring at the rear focal plane of the objective waveplate, phase shifts introduced by the sample are converted into intensity changes, significantly enhancing the visibility of weakly absorbed features. In addition to Zernike imaging, laboratory X-ray microscopes are also exploring other full-field phase-contrast techniques, such as grating-based Talbot–Lau interferometric imaging, which has proven particularly effective for detecting interfaces, voids, delamination, and cracks in low- k materials [78].

Beyond phase contrast approaches, computed laminography (CL) has become an important technique for inspecting planar semiconductor samples. Unlike conventional CT, which requires a full 360° rotation, laminography tilts the rotation axis relative to the incident beam (see Figure 3d), thereby reducing the effective X-ray path length through large planar samples. CL therefore provides a practical solution for non-destructive, large-area inspection of panel-level or tape-level packaging structures [79,80]. Under synchrotron radiation, laminography can achieve spatial resolutions of tens of nanometers, approaching that of CT [81]. Comparative studies by AMD demonstrate that CL enables full-board imaging of large printed circuit boards (PCBs) and system-in-package (SiP) components [82,83] without destructive sample preparation. Although achievable spatial resolution typically falls short of that of conventional 3D X-ray microscopy (XRM), it remains sufficient to detect critical defects in advanced packaging workflows.

As interconnect density continues to increase, non-destructive inspection of critical structures such as redistribution layers (RDLs) [84], micro-bumps, and through-silicon vias (TSVs) now requires sub-micron to nanometer-scale resolution. 3D X-ray microscopy has successfully resolved line widths and spacings in fan-out RDL structures [85,86], with current demonstrations achieving line-width resolution down to $2\ \mu\text{m}$ [87], enabling the detection of minute open-circuit defects in advanced packaging. Micro-bumps and copper pillars are key interconnect elements in 2.5D/3D integration, wafer-level fan-out packaging, and high-bandwidth memory (HBM). As interconnect pitch continues to shrink and current density increases, their structural integrity and reliability have become critical limiting factors for device performance and lifetime. Owing to their small size and buried nature, non-destructive 3D inspection of micro-bumps is essential for both process development and

failure analysis. From a detection perspective, typical characterization targets include internal voids, interface delamination, cracks, and geometric non-uniformity. High-resolution X-ray CT and X-ray microscopy have been widely applied to visualize these features in three dimensions without requiring sample sectioning. Representative studies demonstrate that laboratory-scale 3D X-ray imaging can resolve voids and discontinuities within solder micro-bumps and Cu pillars at submicron spatial resolution, providing direct insights into defect morphology and spatial distribution within bump arrays [88–90]. Simultaneously, among the various failure mechanisms, electromigration-induced void formation is a particularly critical reliability issue for fine-pitch solder micro-bump interconnects. Three-dimensional X-ray imaging has enabled non-destructive observation of the correlation between void morphology and electromigration-driven degradation [91,92]. Meanwhile, through-silicon vias (TSVs)—another core interconnect technology for 3D integration and advanced packaging—require high-precision, non-destructive inspection to detect defects such as voids, delamination, open-via failures, and stress-induced degradation. Due to their buried nature and high aspect ratio, accurate non-destructive TSV inspection has become a critical requirement for yield control and long-term reliability [5]. High-resolution 3D X-ray microscopy and CT have been widely adopted for non-destructive observation of TSV internal structures, enabling the identification of voiding-related defects inaccessible to optical or surface-sensitive techniques. Representative studies demonstrate that X-ray microscopy reliably detects and localizes voids and through-hole-related defects within TSVs, establishing X-ray CT as a practical tool for TSV defect detection in failure analysis environments [93–95]. Notably, for subtle morphological features such as TSV sidewall profiles, bottom curvature, and silicon-dielectric interfaces, absorption contrast often proves insufficient. As shown in Figure 4, phase-contrast X-ray microtomography significantly enhances sensitivity to these features, enabling precise 3D reconstruction of TSV etch profiles and quantitative statistical analysis of via geometry [96]. These results highlight the advantages of phase-contrast imaging in resolving subtle interface and morphological changes that are difficult to capture with absorption contrast alone.

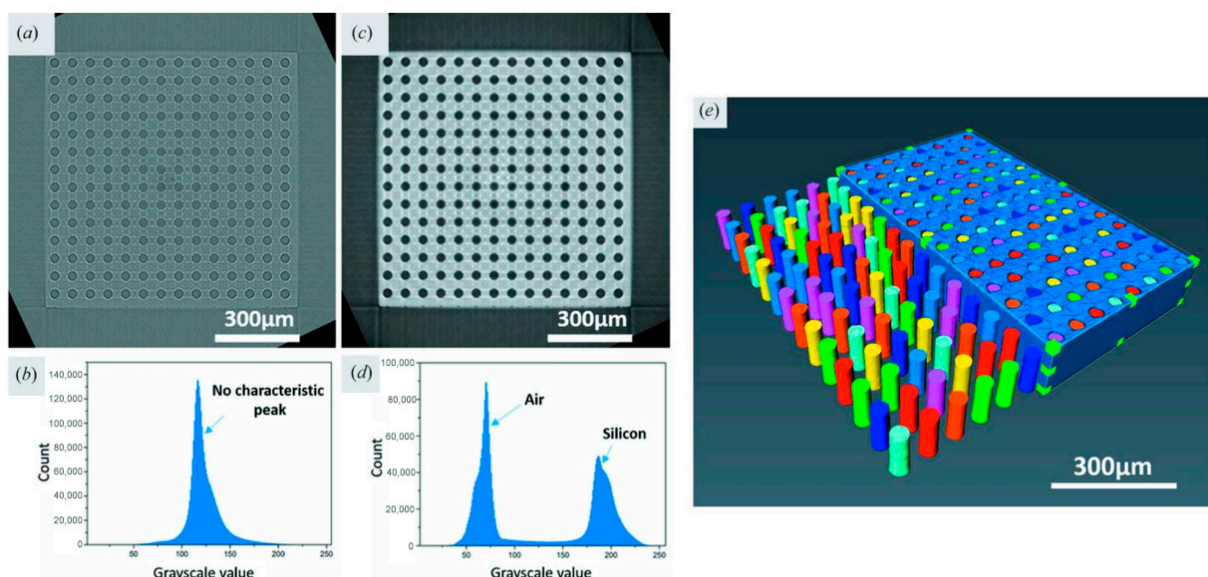


Figure 4. CT reconstruction with and without phase retrieval. (a) A slice reconstructed directly from the unprocessed projections. (b) A histogram of (a). (c) A slice reconstructed from projections with phase retrieval. (d) A histogram of (c). (e) Three-dimensional visualization of TSV etching. Different colors are used to distinguish vias from one another [96]. Reproduced from Li et al. *J. Synchrotron Rad.* 27, 1023–1032 (2020), with permission of the International Union of Crystallography.

Despite significant advances in XCT and XRM technologies, existing commercial XRM systems still cannot simultaneously achieve sub-100-nanometer spatial resolution, high-energy-photon penetration above 15 keV, and meet high-throughput imaging requirements. Regarding resolution alone, current hard X-ray nano-CT imaging, which can approach 20 nanometers, primarily relies on third- and fourth-generation synchrotron radiation sources and has yet to become widespread on laboratory platforms. As packaging technology continues to evolve toward smaller dimensions and higher material homogeneity, the industry increasingly demands imaging methods with enhanced phase sensitivity or novel imaging mechanisms. Against this backdrop, coherent diffraction imaging using synchrotron radiation has emerged as a critical frontier in research. This approach theoretically enables quantitative phase reconstruction at the nanoscale and even the atomic scale, addressing the aforementioned challenges in extreme detection.

3.2. Ptychographic X-Ray Computed Tomography (PXCT) for High-Resolution Three-Dimensional Morphological Reconstruction

In coherent diffraction imaging (CDI), the sample is illuminated by a coherent X-ray beam, and mainly far-field diffraction patterns are recorded without using an imaging lens. These patterns correlate to the sample's Fourier transform but lack phase information. Theoretically, the missing phase can be reconstructed using phase retrieval algorithms, but this typically requires prior knowledge of the sample. For complex, realistic samples, such prior information is often difficult to obtain or insufficient. Ptychography effectively overcomes this limitation by scanning the sample and collecting a series of diffraction patterns from overlapping regions. Figure 3e,f illustrates the differences in the setup between the two methods. This data acquisition method significantly constrains the inverse problem required for reconstruction, thereby substantially reducing dependence on prior information. It only necessitates known scan positions and assumes beam stability during scanning. Simultaneously, the ample overlap of the data enables decoupling the illumination-probe function from the sample's complex amplitude distribution, achieving ultra-high-resolution quantitative phase imaging without requiring high-precision optical components.

Ptychographic X-ray computed tomography (PXCT) represents the first major breakthrough enabling true nanoscale 3D imaging of semiconductor devices. By combining coherent diffraction imaging with tomographic reconstruction, PXCT circumvents the resolution limitations of X-ray optics, achieving resolution primarily determined by the maximum recorded scattering angle. In 2017, Holler et al. reported a landmark demonstration achieving 14.6 nm isotropic 3D resolution in an Intel processor using PXCT [97]. This work employed a focused ion beam (FIB) to prepare cylindrical samples approximately 10 μm in diameter, enabling full-angle rotation during data acquisition. The reconstructed structures clearly resolved copper interconnects, transistor gate structures, and dielectric layers, establishing PXCT as a viable tool for nanoscale semiconductor imaging. Despite its unprecedented resolution, PXCT inherently requires cylindrical specimens to achieve full tomographic rotation. This sample preparation step introduces a degree of destructiveness, limiting PXCT's applicability for routine failure analysis or pinpointing specific defects within large intact chips. These constraints spurred the development of alternative geometries compatible with planar semiconductor samples. To address challenges posed by flat sample geometries and limited angular acquisition, Ptychographic X-ray Laminography Imaging (PyXL) was introduced as a non-destructive alternative to PXCT. In PyXL, the rotation axis is tilted relative to the incident beam, reducing the wedge artifacts that occur when imaging extended planar samples. Holler et al. demonstrated PyXL's feasibility in semiconductor inspection in 2019 by imaging a 16 nm FinFET chip with a spatial resolution of 18.9 nm. Crucially, this approach enables direct imaging of entire chips without columnar sample preparation. Beyond geometric innovation, this work introduces the concept of

macro-to-nano zoom—first utilizing millimeter-scale wide-field-of-view positioning to locate regions of interest, followed by high-resolution in situ scanning of selected areas. This layered workflow significantly enhances the practicality of in situ imaging for failure analysis and reverse engineering tasks [98,99]. Thus, PyXL represents a critical step toward true non-destructive chip inspection, bridging the gap between laboratory demonstrations and application-driven semiconductor analysis.

Although PXCT and PyXL have demonstrated the feasibility of nanoscale 3D imaging, experimental stability remains a major bottleneck, particularly during extended acquisition times required for advanced nodes. Mechanical vibrations, thermal drift, and beam instability severely limit the resolution of conventional tomographic scanning. To overcome these challenges, Aidukas et al. introduced burst ptychography in 2024 [68]. This technique employs rapid sequences of short-exposure diffraction measurements to effectively “freeze” sample motion, followed by computational correction of residual drift. Combined with tomographic backpropagation reconstruction, burst ptychography achieved 4 nm half-width resolution on a commercial 7 nm node logic chip. Individual silicon wafers and metal gate structures are clearly resolved in three dimensions, marking the first demonstration of X-ray imaging technology capable of addressing cutting-edge semiconductor process nodes. Furthermore, this technique enables a 170-fold increase in acquisition rate, from several thousand to 14,000 voxels per second. This achievement represents a qualitative leap in X-ray semiconductor metrology, positioning ptychography as a competitive non-destructive alternative for high-resolution and full-field reconstruction tasks—particularly where 3D information and maximum sample preservation are required.

Beyond the aforementioned technology, to overcome thin-sample limitations, enhance 3D resolution and quantitative imaging capabilities, and maintain manageable computational demands, Shimomura et al. introduced precession X-ray ptychography—a method inspired by precession diffraction. By systematically tilting the incident beam and acquiring multi-angle ptychographic diffraction data, this approach significantly increases angular diversity within datasets. Combined with multilayer reconstruction algorithms, it enables numerical separation of signals from different depth layers. As shown in Figure 5, researchers successfully resolved two circuit layers with a vertical spacing of only 1.4 μm , achieving a spatial resolution of approximately 10 nanometers [100].

Despite its exceptionally high spatial resolution, ptychographic imaging remains limited in widespread commercial chip inspection applications due to its lengthy data acquisition times, high radiation exposure, and substantial computational resource demands. In recent years, rapid advances in algorithms and deep learning methods have enabled significant improvements in the efficiency and robustness of ptychographic imaging. These approaches aim to accelerate data acquisition, reduce radiation exposure, and minimize radiation damage to samples. Current research primarily focuses on reducing the number of projection angles required for reconstruction. Kang et al. proposed the Attentional Ptychographic Task (APT), a physics-based machine learning framework incorporating axial self-attention mechanisms [101]. By leveraging correlations along the rotation axis, APT achieves high-quality 3D reconstructions using only about one-twelfth of the projection angles required by conventional methods, significantly shortening data acquisition time. Beyond angle sparsification, learning-based approaches have also been applied to reconstructions with extremely limited angles. In the same group’s work, Kang et al. demonstrated that transformer-based networks can reconstruct 3D IC structures from single-angle or sparse-angle predictions by leveraging inherent strong structural priors in integrated circuits, such as hierarchical Manhattan geometry [102]. Furthermore, the application of self-supervised learning and pre-training strategies enables networks to optimize ptychographic reconstruction without extensive labeled data, further enhanc-

ing algorithmic robustness [103–106]. These significant algorithmic advancements are propelling ptychographic imaging toward becoming a practical tool for reconstructing high-resolution structures in advanced packaging. While maintaining ultra-high spatial resolution, the technique has achieved notable improvements in imaging speed, dose efficiency, and system robustness.

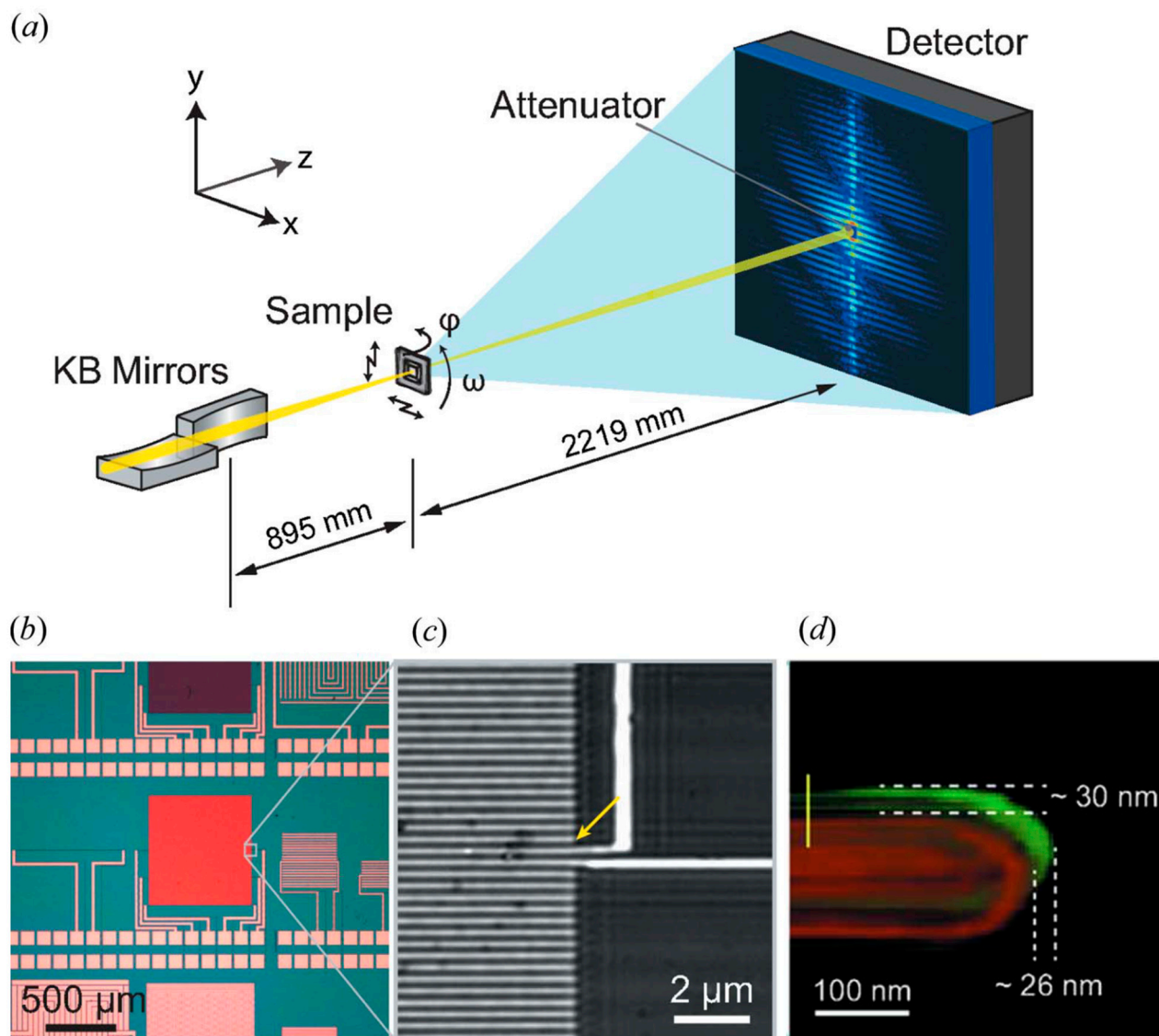


Figure 5. Precession X-ray ptychography for depth-resolved imaging of stacked circuits. (a) Schematic of precession X-ray ptychography experiment. The sample was composed of two circuits, which were stacked with a gap of ~ 1.4 mm. The incident X-ray angle was controlled by adjusting a two-axis rotational stage, where φ and ω are the horizontal and vertical tilt angles, respectively (The arrows indicate the translation (zigzag arrow) and tilt (curve arrow) scanning of the sample). (b,c) Top views of the sample and (c) the X-ray-irradiated position is indicated by the yellow arrow. (d) Reconstructed phase images of the two-layered (red and green) circuit by precession 3PIE [100]. Adapted from Shimomura et al. *Acta Cryst. A* 74, 66–70 (2018) with permission of the International Union of Crystallography.

When the primary objective is three-dimensional reconstruction of buried structures, the most useful comparison is by imaging capability rather than by device application. Table 4, therefore, sorts X-ray imaging approaches discussed in this review.

Table 4. Capability comparison of representative see-through X-ray imaging methods.

Metric	CT	CL	TXM	PXCT/PyXL	Multislice Ptychography
Typical resolution	μm to sub-μm	μm to sub-μm	~30–50 nm	~10 nm	~10 nm
sample compatibility	high	mainly for planar samples	moderate	moderate, more demanding	moderate, favorable for layered planar samples
Contrast mechanism	mainly absorption	absorption	absorption/phase	phase retrieval	phase reconstruction with depth separation
Dose burden	low	moderate	moderate	high	high
maturity	highest	high	moderate to high	research-stage	research-stage
Best-suited role	routine buried-defect inspection	flat-chip/package 3D inspection	higher-resolution local 3D imaging	ultimate nanoscale non-destructive 3D reconstruction	nanoscale layer separation

This comparison shows that no single 3D X-ray imaging mode simultaneously maximizes resolution, penetration, robustness, and throughput, which explains why different scenarios call for different mainstream technologies.

4. X-Ray Spectroscopy for Compositional and Chemical Analysis

Even direct structural reconstruction is often insufficient to explain why a semiconductor interface degrades, why a defect forms, or why device performance shifts during processing and operation. In many cases, the decisive factors are chemical rather than purely geometric: trace metal contamination, dopant redistribution, interdiffusion, oxidation, bonding-state evolution, and interface dipoles can all alter stress, defect formation, carrier transport, and interfacial stability. These are not fully determined by lattice distortion or morphology alone. What is physically required is access to elemental distribution and chemical-state information, especially at buried or ultrathin interfaces. X-ray fluorescence and photoelectron spectroscopy, therefore, become essential when the scientific question is not only what structure is present, but what chemical origin produced that structure, and how composition and bonding constrain its behavior.

4.1. X-Ray Fluorescence (XRF) for Elemental Composition and Contamination Control

XRF analysis is based on the photoelectric effect and the subsequent relaxation process of inner-shell electrons in atoms. When the photon energy exceeds the binding energy of an inner-shell electron, the electron is ejected, leaving a core-level vacancy. This vacancy is subsequently filled by an outer-shell electron, accompanied by the emission of a characteristic X-ray photon with an energy determined by the difference between the two electronic levels. By measuring the energy and intensity of these characteristic photons with detectors such as proportional counters or silicon drift detectors (SDD), both qualitative and quantitative elemental analysis can be achieved. More importantly for semiconductor characterization, XRF can be configured so that the measured fluorescence signal encodes not only elemental presence, but also lateral distribution, surface concentration, or even depth-dependent composition. Hence, XRF is not merely an elemental assay tool, but a route to reconstructing elemental distributions in semiconductor structures.

Current advances in XRF technology primarily focus on improving detection sensitivity and expanding spatial analysis capability. Regarding sensitivity, synchrotron-based microbeam XRF (μ -XRF) has pushed detection limits to the level of hundreds of atoms (see Figure 6a). For instance, a benchmark study by Masteghin et al. demonstrated that XRF microscopy using high-brightness synchrotron nano-beams can detect a few thousand Ga

atoms within a one-second acquisition time, with the detection limit further reduced to several hundred atoms through longer integration [107]. These developments highlight XRF's capability for nanoscale elemental analysis. In parallel with improvements in sensitivity, advances in imaging modes have expanded the accessible field of view. Full-field XRF imaging technology has emerged as a powerful approach for rapid mapping of elemental distribution across centimeter-scale regions while maintaining micrometer-level spatial resolution [108–110]. Together, these developments significantly broaden the applicability of XRF, from a point-analysis method into a multiscale imaging approach for chemical heterogeneity.

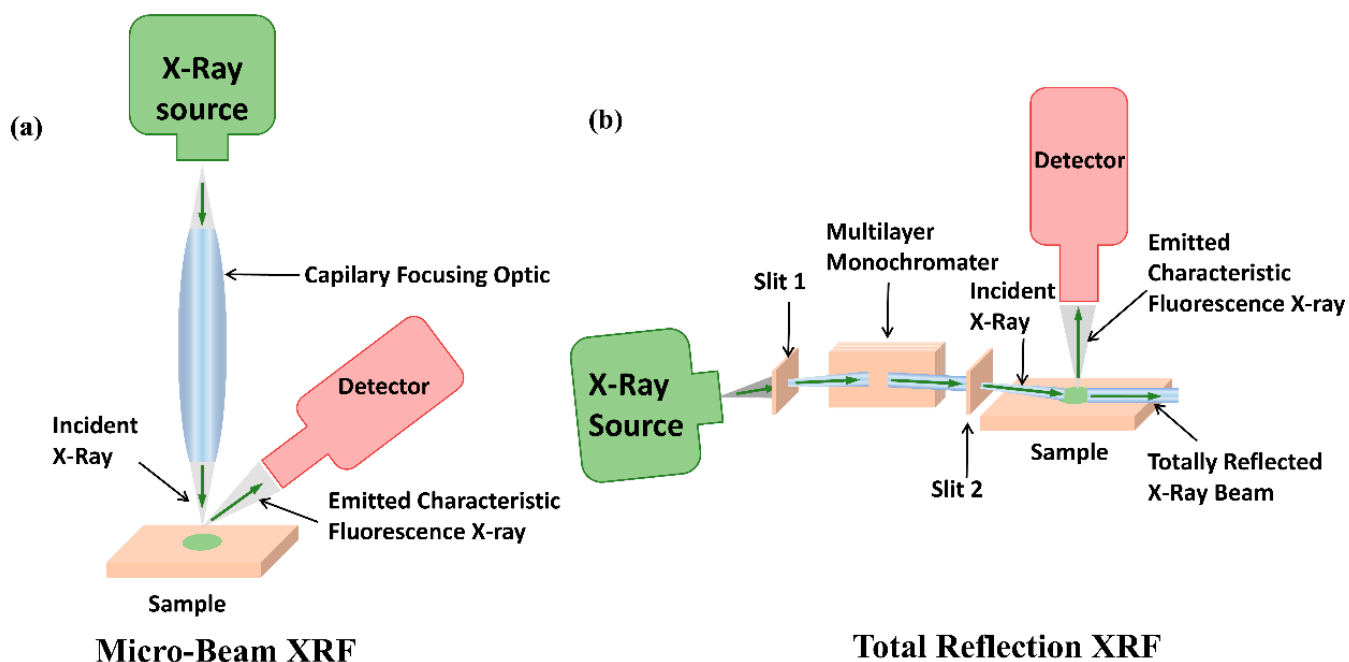


Figure 6. Two typical configurations of XRF. (a) μ -XRF. (b) TXRF. The green arrow indicates the direction of the beam.

For semiconductor surface and thin-film analysis, grazing-incidence geometries such as TXRF and GIXRF provide additional advantages by enhancing surface sensitivity and suppressing substrate background. In industrial applications, these techniques have become the primary methods for balancing high spatial resolution, field-of-view coverage, and measurement repeatability. Both approaches employ grazing-incidence optical paths that not only reduce substrate scattering but also generate a controllable X-ray standing-wave field (XSW) in thin films or multilayer systems, thereby further improving depth sensitivity.

Total reflection X-ray fluorescence (TXRF) operates under the condition of total external reflection when the incident angle is below the critical angle of the substrate (see Figure 6b). Under this geometry, penetration of the excitation beam into the substrate is significantly suppressed, markedly reducing background noise and enhancing sensitivity to surface and near-surface elements. Consequently, TXRF has become the standard technique for monitoring metal surface contamination on silicon wafers. To meet the requirements of industrial in-line monitoring, compact and low-power TXRF spectrometers have been developed, facilitating integration into manufacturing environments and enabling rapid and cost-effective contamination monitoring [111,112]. At the same time, TXRF also illustrates a key principle of chemically sensitive reconstruction: the measurement geometry defines the information depth. Because TXRF is highly surface-selective, it is highly effective for reconstructing near-surface contamination profiles, but its applicability becomes limited when surface roughness increases. Yim et al. systematically investigated the influence of

surface roughness on TXRF signals and found that when the surface root-mean-square roughness exceeds approximately 3 nm, the scattering background increases markedly, fluorescence peaks broaden, and detection limits deteriorate, thereby constraining the reliability of TXRF in realistic advanced-manufacturing conditions [113]. Thus, TXRF is best understood as a highly sensitive but geometrically constrained route for reconstructing elemental distributions at or very near the surface.

A more structurally informative XRF mode is grazing-incidence X-ray fluorescence (GIXRF), especially when combined with X-ray reflectivity (XRR). By precisely controlling the incident angle near the critical angle, GIXRF enables X-rays to generate a standing-wave field within the sample. Because elemental fluorescence is modulated by this field, the angular dependence of the fluorescence signal encodes the vertical distribution of elements. In other words, GIXRF transforms fluorescence intensity into a depth-sensitive signal and thereby enables reconstruction of elemental concentration profiles in thin films and multilayers. Rotella et al. demonstrated this principle through combined GIXRF–XRR analysis, showing that elemental depth profiles can be reconstructed with nanometer-scale resolution when XRR provides constraints on thickness, density, and interfacial roughness [114]. This coupling is important because GIXRF alone is not simply a direct measurement of depth, but an inversion problem whose stability depends on an accurate structural model. Subsequently, the reference-free analysis method [115] further reinforces the point that chemically sensitive X-ray characterization becomes substantially more powerful when tied to accurate structural constraints.

Taken together, XRF-based methods contribute to semiconductor characterization through three complementary reconstruction routes: μ -XRF and full-field XRF reconstruct lateral elemental heterogeneity across different spatial scales; TXRF reconstructs surface and near-surface contamination states with exceptional sensitivity; and GIXRF, especially when combined with XRR, reconstructs depth-dependent composition in thin films and buried interfaces. Therefore, it can be seen that, in modern semiconductor characterization systems, XRF should not be regarded as merely a chemical assay technique, but as an elemental-distribution reconstruction tool that provides the compositional dimension needed to interpret structural evolution, interface quality, and process-induced heterogeneity in complex semiconductor systems. At the same time, the capabilities of XRF are defined by its underlying physical mechanism. XRF provides robust and quantitative information on total elemental composition over relatively large information depths, typically ranging from micrometers to millimeters, which makes it particularly suitable for bulk-sensitive analysis and process monitoring. Its experimental implementation is comparatively straightforward, enabling high-throughput and in-line applications. However, because XRF detects characteristic fluorescence energies that are only weakly affected by chemical environment, it cannot resolve different oxidation states or bonding configurations of the same element. In addition, the relatively large penetration depth makes XRF less suitable for ultrathin surface-specific analysis.

4.2. X-Ray Photoelectron Spectroscopy (XPS) for Chemical Bonding and Surface Analysis

X-ray Photoelectron Spectroscopy (XPS) is based on the photoelectric effect. By measuring the kinetic energy of photoelectrons emitted when materials are excited by X-rays of known energies, XPS enables qualitative and quantitative analysis of the elemental composition, chemical states, and electronic structures at material surfaces. Its typical penetration depth ranges from 1 to 10 nanometers, making it highly sensitive to surface and interfacial phenomena. As semiconductor technology advances toward nanoscale dimensions, heterogeneous integration, and complex three-dimensional structures, XPS has evolved from a traditional surface chemical analysis tool into a versatile characterization platform

capable of resolving band alignment, buried interface chemistry, and depth-dependent elemental distributions. In addition, because core-level shifts and valence-band features respond sensitively to local potential, band bending, and interfacial dipoles, XPS can also reconstruct aspects of buried electronic structure that are inaccessible to purely structural or morphological measurements.

Synchrotron radiation has played a pivotal role in expanding XPS capabilities through its tunable photon energy, high brightness, and polarization control. Particularly crucial for semiconductor applications are beamlines integrated into in situ sample environments, enabling immediate measurement of chemical states and band alignment following thin-film growth or process steps [116–118]. Such in situ measurements effectively eliminate artifacts caused by surface contamination and are essential for reliable interface analysis in oxide, nitride, and wide-bandgap semiconductor systems. For example, the Micro-SX-ARPES beamline at SPring-8 employs a Kirkpatrick–Baez (KB) lens to focus soft X-rays into micrometer-sized spots. This approach overcomes sample size limitations and enables the investigation of the three-dimensional band structure of minute single crystals, particularly dispersion along the vertical direction [118,119]. In parallel with synchrotron developments, laboratory XPS instruments have achieved notable progress, particularly in excitation energy capabilities. The adoption of monochromatized Cr K α sources and liquid-metal-jet Ga K α sources has enabled hard X-ray photoelectron spectroscopy (HAXPES) measurements under laboratory conditions [120,121]. Compared to traditional Al K α excitation, HAXPES achieves an information depth of 20–30 nm, enabling non-destructive chemical state analysis of buried interfaces. The integration of soft and hard X-ray excitation sources in a dual-source configuration further facilitates complementary characterization across different information depths.

One important reconstruction task enabled by XPS is interface electronic-structure analysis. In heterojunctions and metal–semiconductor contacts, device behavior is strongly governed by valence-band offsets, conduction-band offsets, and interfacial charge redistribution. XPS provides a direct and quantitative route for determining such parameters through core-level-to-valence-band analysis, typically using the Kraut method. Band bending induced by interfacial charge transfer causes systematic shifts and broadening of core levels, transforming the spectral lines themselves into probes of localized potential distributions. In situ XPS is particularly crucial as it avoids interference from surface contamination on energy level positions, thereby revealing the true interfacial charge rearrangement and dipole formation [122]. This method has been widely applied to two-dimensional material heterostructures and mixed-dimensional semiconductor systems, distinguishing between Type I and Type II band alignments [123]. It can even resolve approximately 0.2 eV energy level shifts induced by interfacial dipoles, directly reflecting charge transfer across interfaces and the resulting band bending [124].

While band alignment analysis focuses on electronic structure at interfaces, characterizing buried interfaces and the vertical chemical distribution in multilayer films, three-dimensional structures require greater depth-resolved techniques. The detection depth of conventional XPS is typically limited to a few nanometers near the surface. To overcome this limitation, several depth-resolved techniques have been developed. First, Hard X-ray Photoelectron Spectroscopy (HAXPES) employs X-rays with higher photon energies, extending the detection depth to tens of nanometers. This enables HAXPES to non-destructively probe deeply buried interfaces covered by top layers, such as high-k dielectric layers beneath metal gates or device structures beneath encapsulation layers. Budri et al. demonstrate the powerful utility of laboratory-scale HAXPES technology. In analyzing a contact-resistance failure, researchers used high-energy X-rays to penetrate thick metal electrode layers and directly perform chemical-state analysis of the deeply

buried Ti/Al contact interface. The HAXPES spectrum clearly revealed an anomalous fluorine residue at the interface. This residue, originating from a prior process step, formed a high-resistance layer at the interface, ultimately causing device failure. This case highlights HAXPES's irreplaceable value in precisely locating chemical-composition defects within complex integrated-circuit stacks and in resolving practical manufacturing challenges [125].

Another significant technological direction is standing-wave X-ray photoelectron spectroscopy (SW-XPS). By precisely adjusting the wavelength or the angle of incidence of X-rays, the positions of standing-wave antinodes and nodes along the sample depth can be controlled. This enables selective enhancement of photoelectron signals from specific atomic layers, achieving depth-selective analysis at the single-atom-layer level. In extreme ultraviolet (EUV) lithography—a cutting-edge semiconductor manufacturing process—the vertical distribution of photoresist components critically determines performance. Conti et al. employed SW-XPS to perform sub-nanometer depth profiling of EUV photoresist films. By generating standing waves via Bragg reflection and combining angle-scanning, they achieved high-precision quantitative characterization of film-interface roughness, molecular orientation, and interlayer diffusion. By tuning the X-ray standing-wave field, they revealed an interface roughness of only 0.2 nm between the SAM film and the aluminum oxide substrate (see Figure 7). This study experimentally demonstrated, for the first time, the chemical heterogeneity present at the bottom of photoresist films, providing unprecedented insights into their development mechanisms and optimizing performance [126,127].

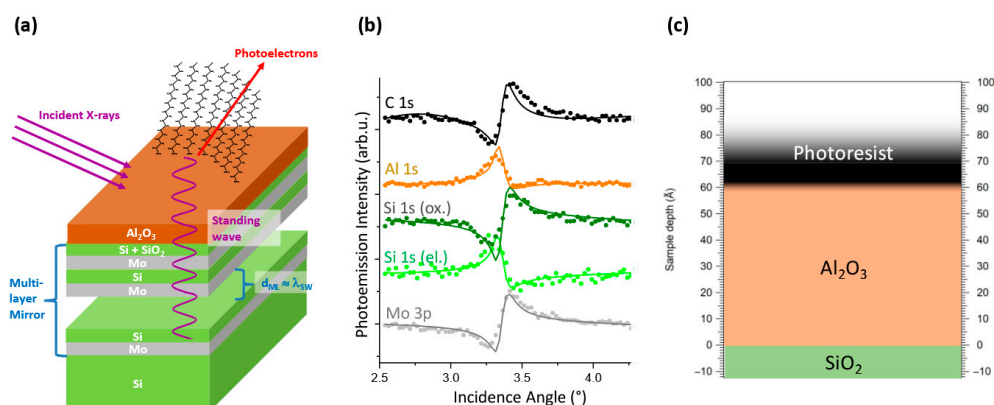


Figure 7. Standing-wave XPS (SW-XPS) for sub-nm depth profiling of EUV photoresist stacks. (a) Experimental setup: a standing wave is generated by Bragg reflection from a multilayer mirror; varying the incident angle shifts the wave antinodes vertically through the sample. (b) Experimental and simulated rocking curves for an e-beam exposed self-assembled monolayer (SAM). (c) Concentration depth profiles reveal an interface roughness of only 0.2 nm between the SAM and Al₂O₃ substrate, and quantify interdiffusion lengths, demonstrating SW-XPS's unique capability for non-destructive chemical depth profiling at the atomic scale [126]. Reproduced from Conti et al. [126], Proc. SPIE 11517, 115170B (2020), with permission from SPIE and the authors.

In summary, XPS is a chemically and electronically meaningful interface state analysis technique. Conventional XPS resolves surface bonding environments and band alignment, HAXPES extends this reconstruction capability to buried stacks, and SW-XPS adds depth selectivity at the sub-nanometer scale. However, XPS-based techniques are intrinsically constrained by trade-offs among information depth, resolution, and experimental efficiency. Conventional XPS provides excellent surface sensitivity at the expense of penetration depth. HAXPES partially alleviates this limitation but is accompanied by reduced signal intensity and higher experimental cost. SW-XPS achieves depth resolution yet requires highly controlled sample preparation and involves substantially increased complexity in data interpretation.

As semiconductor thin films have become increasingly complex, XRF and XPS techniques have evolved from providing simple cross-validation of surface or ultrathin layer elemental content [128] to forming a highly complementary relationship [7,129,130]. XRF delivers bulk or overall elemental distribution, while XPS provides surface-sensitive chemical-state and electronic-structure information. Moreover, XRR provides vertical structural information, including thickness and interface roughness, whereas GISAXS gives lateral structural insights at the nanoscale. Looking back at film metrology mentioned above, thin film and multilayer challenges rarely depend on a single observable. Table 5, therefore, compares how complementary X-ray methods address thickness, roughness, elemental depth distribution, and interface chemistry in stacked semiconductor films.

Table 5. Comparison of X-ray methods for film and near-surface distribution characterization.

Technique	Application	Strength	Limitation
XRR	Thickness, density, interface roughness	Sub-nm roughness	Model-dependent; limited lateral information
GI-SAXS	Surface morphology, lateral roughness, nanostructure shape	In-plane morphology and nanoscale correlation	Less direct for absolute thickness
μ -XRF	Elemental analysis	Quantitative; lateral compositional mapping over large areas, μm -mm depth	Depth sensitivity indirect
GIXRF	Elemental depth distribution	Element-specific depth sensitivity	Requires modeling and usually structural constraints
XPS	Surface chemistry, bonding state, band alignment	Direct sensitivity to chemical-state and electronic structure	Only a few nm information depth
HAXPES/SW-XPS	Buried interface chemistry or depth-selective analysis	Extends XPS to buried or sub-nm depth resolution	More specialized and experimentally demanding

5. Conclusions and Perspective

In this review, a systematic overview of X-ray characterization techniques for advanced semiconductor structures has been presented, covering diffraction-based methods, small-angle scattering, computed tomography, and spectroscopic approaches. These techniques provide complementary insights into structural, morphological, and compositional properties across multiple length scales, making them essential tools for addressing the increasing complexity of modern semiconductor devices and advanced packaging systems.

The comparison of different methods highlights that no single technique is sufficient to fully resolve the diverse structural challenges encountered in practice, such as high-aspect-ratio features, buried interfaces, and wafer architectures. Instead, effective characterization often relies on combining multiple X-ray modalities, each contributing distinct information and constraints. In this context, understanding the capabilities and limitations of each technique is critical for selecting appropriate measurement strategies and for interpreting experimental results in a consistent manner.

It is also important to recognize that many X-ray characterization techniques rely on model-based interpretation, where structural information is inferred from measured signals rather than directly observed. From this perspective, different X-ray methods can be viewed as probing the material system through different physical encoding mechanisms, including reciprocal-space scattering, real-space projection, and element-specific emission or excitation processes. The extraction of meaningful structural parameters, therefore,

depends not only on data quality but also on the underlying assumptions, reconstruction models, and the stability of the associated inverse problems.

At present, however, no single X-ray technique can simultaneously satisfy the demands for nanoscale resolution, large penetration depth, high throughput, chemical sensitivity, and industrial compatibility. Synchrotron-based methods continue to define the frontier of high-resolution and multidimensional characterization, while laboratory systems remain more attractive for routine process monitoring and wider technological adoption. This gap defines the central challenge for the next stage of X-ray metrology development.

Looking forward, future improvements will focus on better light sources, robust algorithms, and the integration of multiple characterization techniques into unified systems.

First, while current high-brightness synchrotron radiation sources provide vital support for integrated circuit metrology, their operational costs and accessibility constraints limit widespread industrial adoption. Future developments in laboratory microfocus sources, advanced optical systems, and high-dynamic-range detectors will further enhance spatial resolution and data acquisition efficiency in laboratory setups. This will drive X-ray technology toward high stability, repeatability, and online acquisition capabilities, bridging the gap between research facilities and industrial applications.

To better contextualize the role of X-ray source development across different techniques, Table 6 summarizes the typical radiation source characteristics, including radiation requirements (monochromatic, polychromatic, or coherent), energy ranges, and beam properties, together with their implications for radiation damage.

Most diffraction- and spectroscopy-based techniques (e.g., XRD, XRR, SAXS, and XPS) rely on monochromatic radiation, whereas imaging approaches such as CT can operate with polychromatic sources. In contrast, CDI and ptychography primarily depend on beam coherence. While both laboratory and synchrotron sources are used, the higher photon flux and smaller beam sizes at synchrotron facilities generally increase the risk of radiation damage. These differences lead to technique-specific source requirements. For example, SAXS and HAXPES require both sufficient photon energy for penetration and strict monochromaticity, placing high demands on photon flux. As a result, they still rely heavily on synchrotron radiation. In contrast, CDI and ptychography require highly coherent and intense beams, making laboratory-scale implementation even more challenging. Overall, laboratory sources are emerging as a key direction for development. Promising laboratory light sources currently include liquid metal jet sources and inverse Compton scattering sources. Liquid metal jet sources provide high brightness and flexible energy selection for high-resolution scattering and diffraction measurements, although their spatial coherence remains lower than that of synchrotron radiation. Inverse Compton scattering sources offer the potential for compact and high-brightness hard X-ray generation, although their technical maturity and system stability are still under active development [131,132]. Advancements in these sources will enable imaging techniques centered on ptychography—such as Bragg ptychography and ptychographic X-ray computed tomography—to achieve both high resolution and large field-of-view. Such developments represent an important pathway toward nanoscale three-dimensional characterization of semiconductor devices.

Secondly, the structural reconstruction of scattering and diffraction data typically relies on complex model fitting and iterative computations, demanding high algorithmic efficiency and stability. Specific applications of deep learning have been discussed in the respective X-ray technique sections. Overall, four key directions are emerging for AI-assisted X-ray analysis: physics-constrained deep learning frameworks enable rapid and robust reconstruction under sparse data conditions; data fusion strategies integrate results from hybrid metrology to enhance prediction accuracy [133]; adaptive scanning strategies driven by real-time feedback to improve data acquisition efficiency; and automated defect

recognition for high-throughput wafer inspection. Furthermore, AI-assisted data analysis and defect identification tools can rapidly flag anomalous regions in high-throughput wafer inspection. Crucially, future applications may extend to adjusting process parameters and comprehending complex structures. Through these approaches, AI not only accelerates and improves the reliability of structural reconstruction but also reduces reliance on ultra-high-coherence light sources or on prolonged exposure times, thereby enabling faster and more robust semiconductor inspection workflows.

Table 6. X-ray source characteristics and radiation for different techniques.

Technique	Source Type	Radiation Requirement	Energy/Tube Voltage	Beam Size	Radiation Damage Consideration
XRD	Laboratory & Synchrotron	Monochromatic	~8 keV (mainly Cu K α 1)	mm- μ m (lab); sub- μ m (sync)	Generally low, Higher for scanning and white light measurements
SAXS	Laboratory (LMJ & rotating anode) & Synchrotron	Monochromatic	8–17 keV (Cu/Mo/W K α 1)	μ m	Higher flux and energy typically required
XRR	Laboratory & Synchrotron	Monochromatic	~8 keV (mainly Cu K α 1)	μ m	Grazing-incidence geometry more radiation-tolerant
XRF	Mainly Laboratory	Typically, polychromatic or filtered	~50 kV (tube voltage)	mm	Generally low
XPS	Laboratory (LMJ) & Synchrotron	Monochromatic	~1.49 keV (mainly Al K α 1) higher for HAXPES (Cr K α 1, Ga K α 1, SR)	mm- μ m	Surface damage possible
CT/TXM	Mainly Laboratory	Polychromatic	up to ~130–160 kV (tube voltage)	μ m-nm	Generally low, higher in high-resolution scans
PXCT	Synchrotron (coherent beam)	Monochromatic (coherence more critical)	5–10 keV (typically 6)	nm	High dose due to focused beam and long scanning acquisition

Finally, individual X-ray techniques are typically sensitive to specific structural parameters but cannot simultaneously capture geometric, compositional, and strain information. As semiconductor devices enter the sub-10-nanometer scale, structural features exhibit multiscale, multi-material, and multi-physics coupling characteristics. Therefore, hybrid metrology will become an essential pathway for future structural characterization. Methodologically, future trends extend beyond the combined application of different X-ray techniques to include cross-integration with electron microscopy, optical scattering metrology, electrical characterization, and other approaches. By synthesizing the most reliable information from diverse physical mechanisms to achieve data complementarity and cross-validation, the confidence and stability of measurement results can be significantly enhanced. Currently, established combined modes for thin-film and multilayer structure analysis include XRR with GIXRF [134–136] and XRF with XPS [114,137], enabling simultaneous acquisition of thickness, density, composition, and crystal-structure information. Other combinations include CD-SAXS and SEM [138], OCD, XRF [139], and more. At the system level, future equipment development may increasingly adopt multiplexed architecture designs. Through optical path design and energy switching, rapid switching or simultaneous acquisition of XRD, XRF, CT, and scattering techniques can be achieved. Current explorations include integrated XRF-CT beamline designs [140] and desktop CT systems incorporating XRF modules [141,142].

In summary, with continued advances in laboratory X-ray sources, AI-driven data analysis, and hybrid characterization strategies, X-ray techniques are poised to play an increasingly central role in the study and development of next-generation semiconductor devices and other complex material systems. These developments will support high-resolution,

multi-dimensional structural analysis that is both practical and non-destructive, enabling improved design, process optimization, and quality control in advanced technologies.

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